What you did at home

AI Methods for Science – Part I – Week 2, Thursday

Yonatan Poles **Amit** Tsipin, **Adi** Shecter **Oran** Szachter **Tamar** Avraham

22 October 2026

1. The three failure modes, on 200 numbers

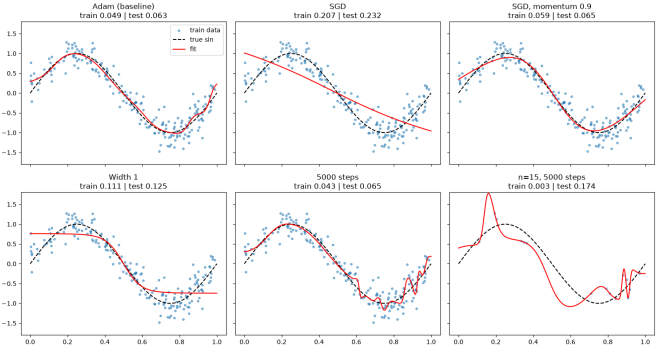


figure: **Oran**. Train / test MSE; the floor on his sample: 0.055 / 0.058 ($\sigma^2 = 0.0625$).

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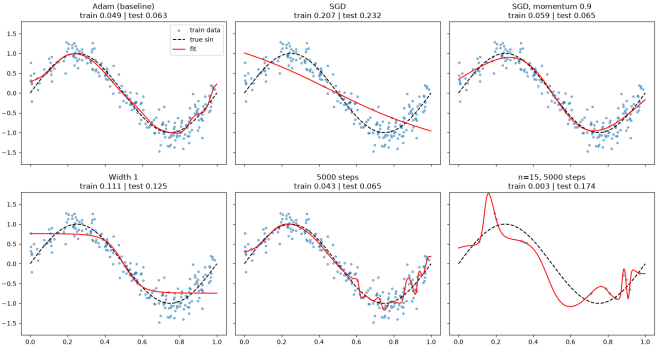


figure: **Oran**. Train / test MSE; the floor on his sample: 0.055 / 0.058 ($\sigma^2 = 0.0625$).

- **width 1**: one tanh cannot bend twice. No good θ exists.
- **SGD**: train MSE 0.21 after 2000 steps; with momentum 0.06, the floor is 0.055. A good θ exists, not reached yet (at $5\times$ the η it is).
- **5000 steps**, $n = 15$: train *below* the floor, test up. Fitting the noise. The floor is the true function's error on the same points: the number to beat, and the one you cannot beat honestly.

2. The noise, and where it comes from

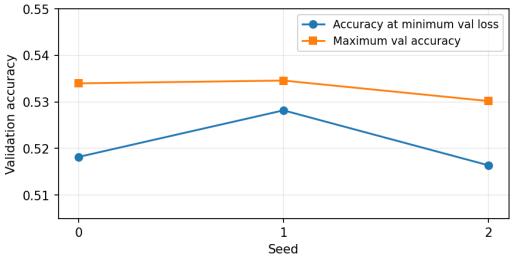


figure: **Yonatan**. E1, three seeds.

`val_acc_at_best`, same code, same seeds

seed	Yonatan	Amit & Adi	Oran	Tamar
0	51.8	51.8	**54.1**	**53.5**
1	52.8	52.8	52.8	52.5
2	51.6	51.6	51.6	50.9

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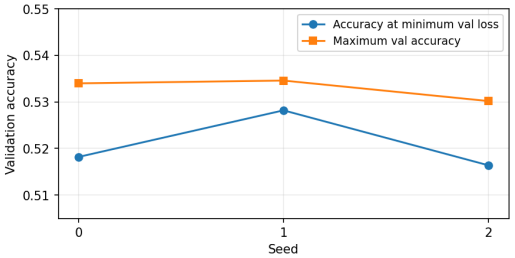


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- seeds: 1–3 points. Machines: the same again. Seed 0 gave 51.8 on two laptops and 54.1 on a third.
- SGD is chaotic: a rounding difference in the last bit grows into a different trajectory, as a new seed would. Nobody could see this inside one report.
- the rule: a difference smaller than the spread is not a result.

3. The learning rate: crawl, work, bounce, blow up

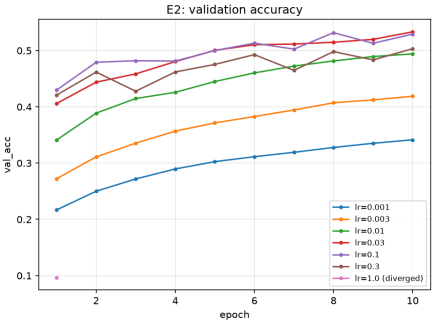


figure: **Amit & Adi**. E2, validation accuracy per epoch.

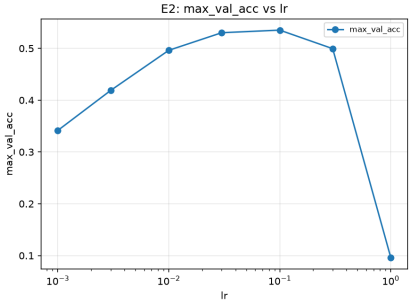


figure: **Tamar**. E2, best validation accuracy vs. η.

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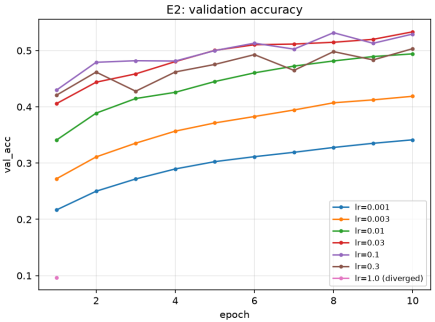


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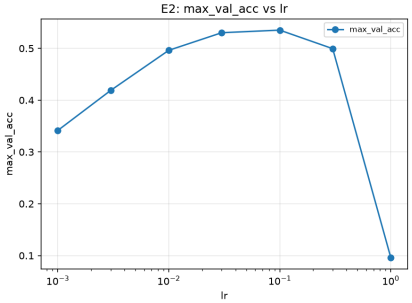


figure: **Tamar**. E2, best validation accuracy vs. η.

- a step converges iff $\eta\lambda < 2$ on the stiffest direction: $\eta = 1$ is `nan` in epoch 1.
- Stability edge is 0.3, the best choice is 3–$10\times$ below it. This is a decent thumb rule.

4. The gap: parameters do not close it, data does

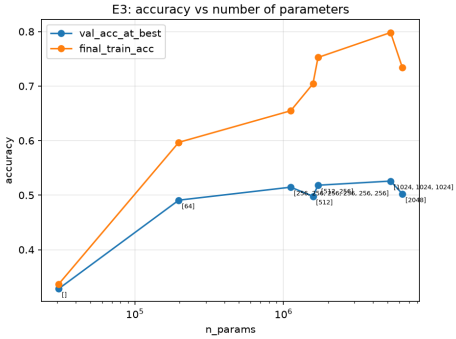


figure: **Amit & Adi**. E3, 15 epochs, seven architectures.

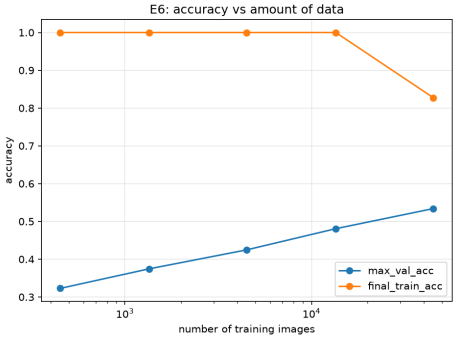


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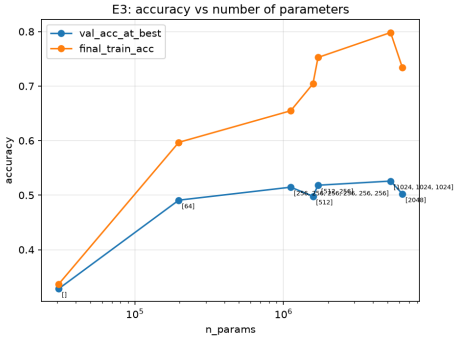


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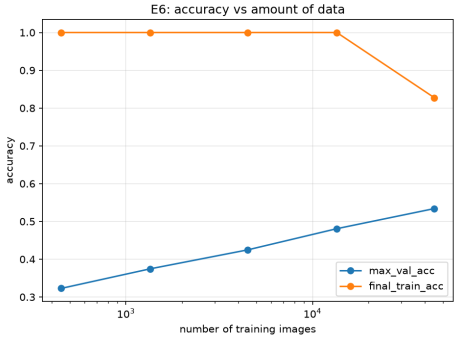


figure: **Amit & Adi**. E6, 450 to $45{,}000$ images.

- left: training $34 \to 80$, validation flat at 50–53 from 1M to 6M parameters. $32\times$ the parameters buy one point.
- right: $+10$ points per decade of data, the gap $68 \to 29$. Capacity is not the knob; data is.
- **Tamar**, three seeds each: [512,256] beats [2048] by 2 points with $4\times$ fewer parameters.

5. Training too long: the loss turns, the accuracy does not

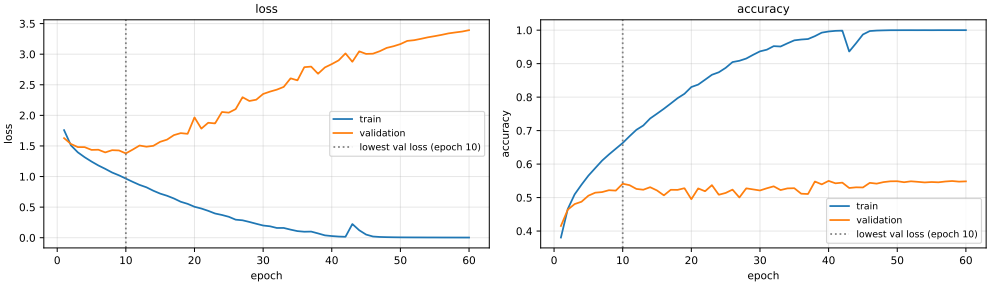


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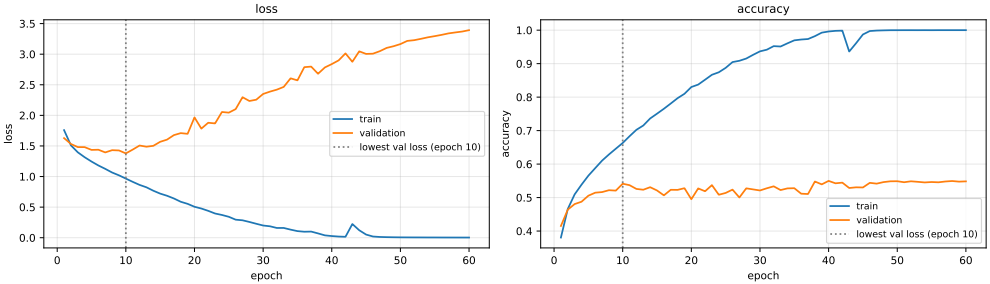


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- after epoch 7–10 the net stops changing its answers and starts believing them. Accuracy counts the argmax: unchanged. Cross-entropy charges $-\log p$ on the wrong ones: $p \to 0.01$ costs 4.6 per mistake, so the loss doubles while the accuracy creeps up to 55.
- at epoch 60: training loss 0.003, training accuracy 100%. The net has memorized $45{,}000$ images. It is not wrong more often; it is wrong with confidence.
- stop on the loss (**Yonatan**, **Tamar**: "how sure the system is matters") or on the accuracy (**Oran**: "the measure we care about"): both fine. Report the number of the epoch you stopped at.

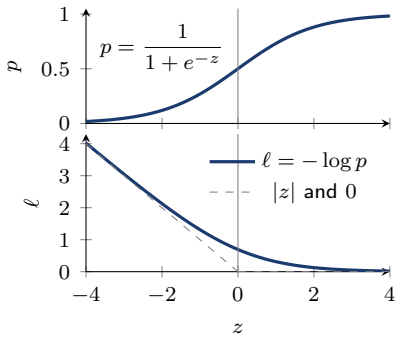
5b. Why the loss keeps rising

Softmax, K classes, true class y:

$$\ell = -\log \frac{e^{f_y}}{\sum_j e^{f_j}} = \log \sum_j e^{f_j - f_y}$$

Two classes, $z = f_y - f_{\text{other}}$: $\ell = \log(1 + e^{-z})$

- $z \gg 1$: right and sure, $\ell \approx e^{-z} \to 0$
- $z \ll -1$: wrong and sure, $\ell \approx |z|$
- accuracy only asks: is $z > 0$?



All training $z_i > 0$. Scale $f \to sf$: training loss $\sum_i e^{-sz_i} \to 0$; a wrong validation point costs $s|z_i|$. Nothing stops s.

6. Batch size: the same run on three x-axes

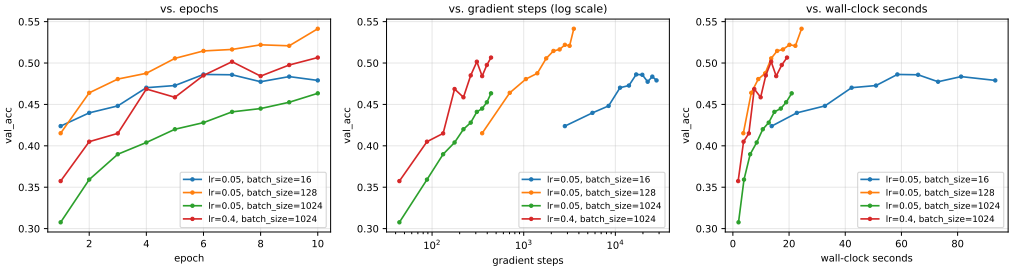


figure: **Oran**. E5, validation accuracy; $B = 16,\ 128,\ 1024$ at $\eta = 0.05$, and $B = 1024$ at $\eta = 0.4$.

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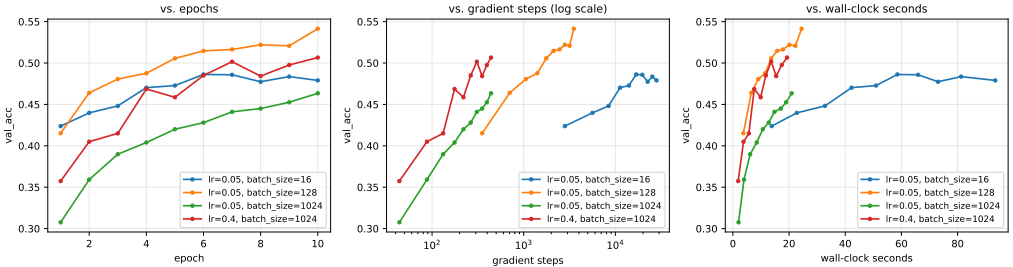


figure: **Oran**. E5, validation accuracy; $B = 16, 128, 1024$ at $\eta = 0.05$, and $B = 1024$ at $\eta = 0.4$.

- per gradient step the big batch wins: each step averages $64\times$ more images, so its gradient is cleaner. Per epoch the small batch wins: 2813 steps against 44. Per second: hardware; $B = 16$ cannot fill the CPU and costs 3–$4\times$ per epoch on every machine.
- the noise of a step $\propto \eta/B$. Raise η with B: $0.4/1024 = 0.05/128$, and $B = 1024$ recovers 5 points (**Oran**, **Tamar** both wrote the ratio down).
- one batch size is not better than another. A pair (η, B) is a point on a line, and the line is what matters.

7. Depth and scale: the gradient is a product

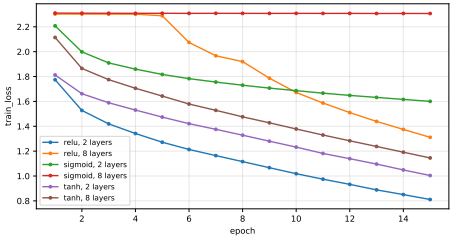


figure: **Oran**. E7, training loss, 2 and 8 layers.

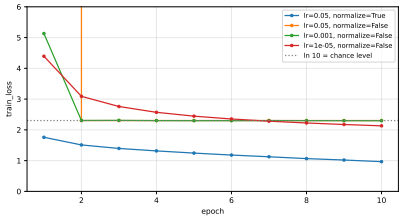


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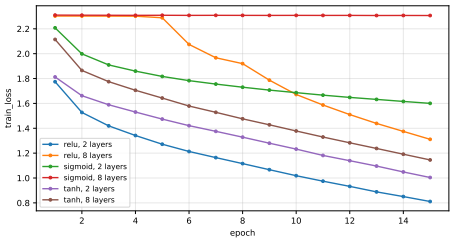


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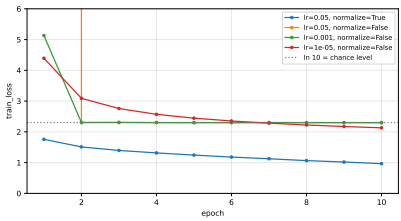


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- backward pass: one factor per layer, multiplied. $\sigma' \leq \frac{1}{4}$, so $4^{-8} \approx 10^{-5}$: the first layers never hear anything, loss at $\ln 10$ for ever. ReLU$\times 8$ waits five epochs (the signal shrinks to the bias floor at the default init), then wakes up.
- E8: the first layer's gradient is $\boldsymbol{\delta}\,\mathbf{x}^\top$; raw pixels are $100\times$ larger, its curvature $10^4\times$. $\eta = 0.05$: 10^{30}, then a net that answers the same for every image ($\ln 10$, not `nan`). $\eta/5000$ is safe for that layer and starves the other seven: 30%.
- one η for all layers needs all layers alike: standardize, or Adam (E12).